



## help binsregselect

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### Title

**binsregselect** — Data-driven IMSE-Optimal Partitioning/Binning Selection for Binscatter.

### Syntax

```
binsregselect depvar indvar [othercovs] [if] [in] [weight] [, deriv(v)
               absorb(absvars) reghdfeopt(reghdfe_option)
               bins(p s) binspos(position) binsmethod(method) nbinsrot(#)
               nbins(nbinsopt)
               pselect(numlist) sselect(numlist)
               simsgrid(#) savegrid(filename) replace
               dfcheck(n1 n2) masspoints(masspointsoption)
               vce(vcetype) usegtools(on/off) precision(single/double) useeffn(#)
               randcut(#)]
```

where *depvar* is the dependent variable, *indvar* is the independent variable for binning, and *othercovs* are other covariates to be controlled for.

The degree of the piecewise polynomial *p*, the number of smoothness constraints *s*, and the derivative order *v* are integers satisfying  $0 \leq s, v \leq p$ , which can take different values in each case.

**fweights**, **aweight**s and **pweight**s are allowed; see [weight](#).

### Description

**binsregselect** implements data-driven procedures for selecting the number of bins for binscatter estimation. The selected number is optimal in minimizing the (asymptotic) integrated mean squared error (IMSE).

### Options

#### Estimand

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**deriv**(*v*) specifies the derivative order of the regression function for estimation, testing and plotting. The default is **deriv**(0), which corresponds to the function itself.

#### Reghdfe

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**absorb**(*absvars*) specifies categorical variables (or interactions) representing the fixed effects to be absorbed. This is equivalent to including an indicator/dummy variable for each category of each *absvar*. When **absorb**() is specified, the community-contributed command **reghdfe** instead of the command **regress** is used.

**reghdfeopt**(*reghdfe\_option*) options to be passed on to **reghdfe**. Important: **absorb**() and **vce**() should not be specified within this option.

For more information about the community-contributed command **reghdfe**, please see <https://scorreia.com/software/reghdfe/>.

#### Binning/Degree/Smoothness Selection

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**bins**(*p s*) sets a piecewise polynomial of degree *p* with *s* smoothness constraints for data-driven (IMSE-optimal) selection of the partitioning/binning scheme. The default is **bins**(0 0), which corresponds to piecewise constant (canonical binscatter).

**binspos**(*position*) specifies the position of binning knots. The default is **binspos(qs)**, which corresponds to quantile-spaced binning (canonical **binscatter**). Other option is **es** for evenly-spaced binning.

**binsmethod**(*method*) specifies the method for data-driven selection of the number of bins. The default is **binsmethod(dpi)**, which corresponds to the IMSE-optimal direct plug-in rule. The other option is: **rot** for rule of thumb implementation.

**nbinsrot**(#) specifies an initial number of bins value used to construct the DPI number of bins selector. If not specified, the data-driven ROT selector is used instead.

**nbins**(*nbinsopt*) sets the number of bins for degree/smoothness selection. If **nbins(T)** is specified, the command selects the number of bins instead, given the specified degree and smoothness. If a *numlist* with more than one number is specified, the command selects the number of bins within this list.

**pselect**(*numlist*) specifies a list of numbers within which the degree of polynomial *p* for point estimation is selected.

**sselect**(*numlist*) specifies a list of numbers within which the number of smoothness constraints *s* for point estimation is selected. If not specified, for each value *p* supplied in the option **pselect()**, only the piecewise polynomial with the maximum smoothness is considered, i.e., *s=p*.

Note: To implement the degree or smoothness selection, in addition to **pselect()** or **sselect()**, **nbins**(#) must be specified.

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#### Evaluation Points Grid Generation

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**simsgrid**(#) specifies the number of evaluation points of an evenly-spaced grid within each bin used for evaluation of the supremum (infimum or *Lp* metric) operation needed to construct confidence bands and hypothesis testing procedures. The default is **simsgrid(20)**, which corresponds to 20 evenly-spaced evaluation points within each bin for approximating the supremum (or infimum) operator.

**savegrid**(*filename*) specifies a filename for storing the simulation grid of evaluation points. It contains the following variables: *indvar*, which is a sequence of evaluation points used in approximation; all control variables in *othercovs*, which take values of zero for prediction purpose; *binsreg\_isknot*, indicating whether the evaluation point is an inner knot; and *binsreg\_bin*, indicating which bin the evaluation point belongs to.

**replace** overwrites the existing file when saving the grid.

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#### Mass Points and Degrees of Freedom

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**dfcheck**(*n1 n2*) sets cutoff values for minimum effective sample size checks, which take into account the number of unique values of *indvar* (i.e., adjusting for the number of mass points), number of clusters, and degrees of freedom of the different statistical models considered. The default is **dfcheck(20 30)**. See Cattaneo, Crump, Farrell and Feng (2025) for more details.

**masspoints**(*masspointsoption*) specifies how mass points in *indvar* are handled. By default, all mass point and degrees of freedom checks are implemented.

Available options:

**masspoints(noadjust)** omits mass point checks and the corresponding effective sample size adjustments.

**masspoints(nolocalcheck)** omits within-bin mass point and degrees of freedom checks.

**masspoints(off)** sets **masspoints(noadjust)** and **masspoints(nolocalcheck)** simultaneously.

**masspoints(veryfew)** forces the command to proceed as if *indvar* has only a few number of mass points (i.e., distinct values). In other words, forces the command to proceed as if the mass point and degrees of freedom checks were failed.

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## Other Options

**vce**(*vcetype*) specifies the *vcetype* for variance estimation used by the command **regress** (or **reghdfe** if **absorb()** is specified). The default is **vce(robust)**.

**usegtools**(*on/off*) forces the use of several commands in the community-distributed Stata package **gtools** to speed the computation up, if *on* is specified. Default is **usegtools(off)**.

For more information about the package **gtools**, please see <https://gtools.readthedocs.io/en/latest/index.html>.

**precision**(*single/double*) controls the storage type used for internally generated Stata variables. The default is **precision(double)**, which uses **double** storage for temporary and generated variables to help align Stata results with the R and Python implementations. Specify **precision(single)** to preserve the legacy Stata implementation using **float** storage for those variables. Stata matrix and Mata computations remain double precision in either case.

**useeffn**(*#*) specifies the effective sample size *#* to be used when computing the (IMSE-optimal) number of bins. This option is useful for extrapolating the optimal number of bins to larger (or smaller) datasets than the one used to compute it.

**randcut**(*#*) specifies the upper bound on a uniformly distributed variable used to draw a subsample for bins selection. Observations for which **runiform()** <=*#* are used. *#* must be between 0 and 1.

## Examples

```
Setup
. sysuse auto
```

```
Select IMSE-optimal number of bins using DPI-procedure
. binsregselect mpg weight foreign
```

## Stored results

### Scalars

|                          |                                                                     |
|--------------------------|---------------------------------------------------------------------|
| <b>e(N)</b>              | number of observations                                              |
| <b>e(Ndist)</b>          | number of distinct values                                           |
| <b>e(Nclust)</b>         | number of clusters                                                  |
| <b>e(deriv)</b>          | order of derivative                                                 |
| <b>e(imse_bsq_rot)</b>   | bias constant in IMSE, ROT selection                                |
| <b>e(imse_var_rot)</b>   | variance constant in IMSE, ROT selection                            |
| <b>e(imse_bsq_dpi)</b>   | bias constant in IMSE, DPI selection                                |
| <b>e(imse_var_dpi)</b>   | variance constant in IMSE, DPI selection                            |
| <b>e(nbinsrot_poly)</b>  | ROT number of bins, unregularized                                   |
| <b>e(nbinsrot_regul)</b> | ROT number of bins, regularized or user-specified                   |
| <b>e(nbinsrot_uknot)</b> | ROT number of bins, unique knots                                    |
| <b>e(nbinsdpi)</b>       | DPI number of bins                                                  |
| <b>e(nbinsdpi_uknot)</b> | DPI number of bins, unique knots                                    |
| <b>e(prot_poly)</b>      | ROT degree of polynomial, unregularized                             |
| <b>e(prot_regul)</b>     | ROT degree of polynomial, regularized or user-specified             |
| <b>e(prot_uknot)</b>     | ROT degree of polynomial, unique knots                              |
| <b>e(pdpi)</b>           | DPI degree of polynomial                                            |
| <b>e(pdpi_uknot)</b>     | DPI degree of polynomial, unique knots                              |
| <b>e(srot_poly)</b>      | ROT number of smoothness constraints, unregularized                 |
| <b>e(srot_regul)</b>     | ROT number of smoothness constraints, regularized or user-specified |
| <b>e(srot_uknot)</b>     | ROT number of smoothness constraints, unique knots                  |
| <b>e(sdpi)</b>           | DPI number of smoothness constraints                                |
| <b>e(sdpi_uknot)</b>     | DPI number of smoothness constraints, unique knots                  |

### Matrices

|                            |                                                                           |
|----------------------------|---------------------------------------------------------------------------|
| <b>e(knot)</b>             | numlist of knots                                                          |
| <b>e(m_p)</b>              | vector of degrees of polynomial                                           |
| <b>e(m_s)</b>              | vector of number of smoothness constraints                                |
| <b>e(m_nbinsrot_poly)</b>  | ROT number of bins, unregularized, for each pair of degree and smoothness |
| <b>e(m_nbinsrot_regul)</b> | ROT number of bins, regularized or user-specified, for                    |

each pair of degree and smoothness  
**e(m\_nbinsrot\_uknot)** ROT number of bins, unique knots, for each pair of degree and smoothness  
**e(m\_nbinsdpi)** DPI number of bins, for each pair of degree and smoothness  
**e(m\_nbinsdpi\_uknot)** DPI number of bins, unique knots, for each pair of degree and smoothness  
**e(m\_imse\_bsq\_rot)** bias constant in IMSE, ROT selection, for each pair of degree and smoothness  
**e(m\_imse\_var\_rot)** variance constant in IMSE, ROT selection, for each pair of degree and smoothness  
**e(m\_imse\_bsq\_dpi)** bias constant in IMSE, DPI selection, for each pair of degree and smoothness  
**e(m\_imse\_var\_dpi)** variance constant in IMSE, DPI selection, for each pair of degree and smoothness

## References

- Cattaneo, M. D., R. K. Crump, M. H. Farrell, and Y. Feng. 2024. [On Binscatter](#). *American Economic Review* 114(5): 1488-1514.
- Supplemental Appendix for On Binscatter. [Supplemental Appendix](#).
- Cattaneo, M. D., R. K. Crump, M. H. Farrell, and Y. Feng. 2026. [Nonlinear Binscatter Methods](#). *Review of Economics and Statistics*, revise and resubmit.
- Supplemental Appendix for Nonlinear Binscatter Methods. [Supplemental Appendix](#).
- Cattaneo, M. D., R. K. Crump, M. H. Farrell, and Y. Feng. 2025. [Binscatter Regressions](#). *Stata Journal* 25(1): 3-50.

## Authors

Matias D. Cattaneo (maintainer). [matias.d.cattaneo@gmail.com](mailto:matias.d.cattaneo@gmail.com).

Richard K. Crump. [richard.crump@gmail.com](mailto:richard.crump@gmail.com).

Max H. Farrell. [mhfarrell@gmail.com](mailto:mhfarrell@gmail.com).

Yingjie Feng. [fengyingjiepk@gmail.com](mailto:fengyingjiepk@gmail.com).